RESONANCE

AC CIRCUITS

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Resonance Frequency

Series Resonance

Parallel Resonance

RESONANCE FREQUENCY

RESONANCE FREQUENCY, f_R

$$f_R = \frac{1}{2\pi\sqrt{LC}}$$

Resultant reactance is zero $[X_L = X_C]$

Circuit behaves like a **pure resistance**

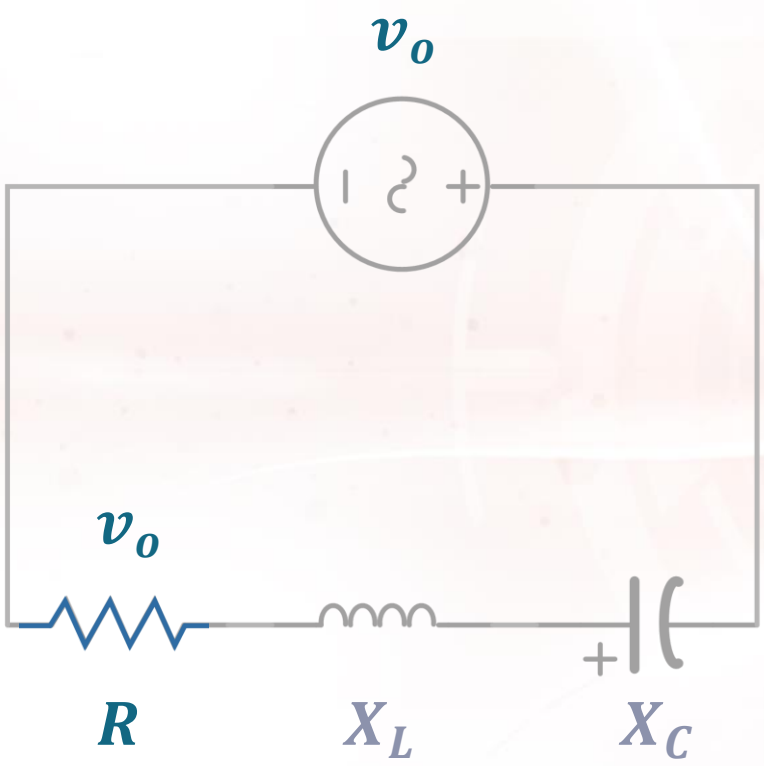
Total current is **in phase** with the impressed voltage

Power factor is **unity**

SERIES RESONANCE

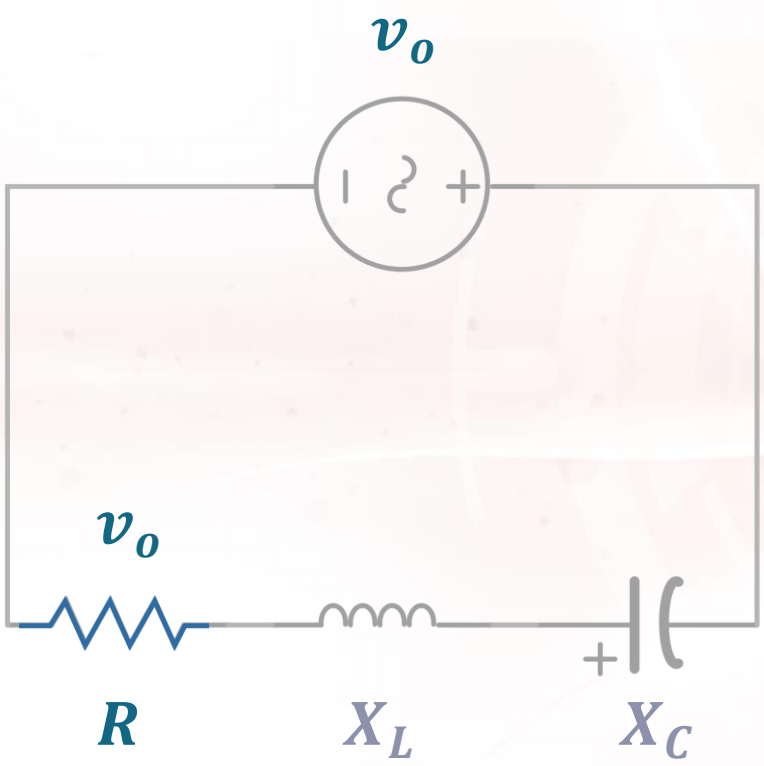
SERIES RESONANCE

$$X_L = X_C$$

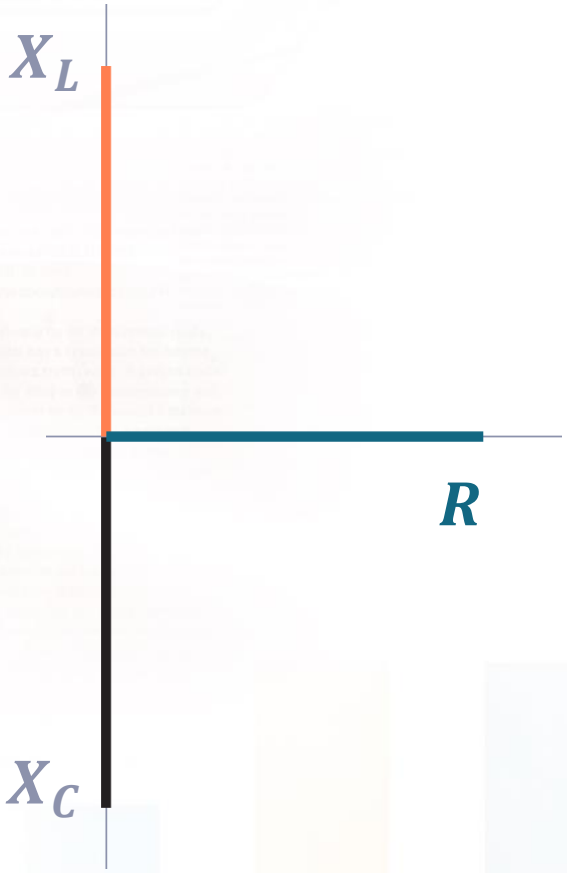


Circuit Diagram

$X_L = X_C$



Circuit Diagram



Impedance Phasor Diagram

A series circuit consists of a **30-ohm** resistor, a **0.104-henry** inductor, and a **40-μf** capacitor. If a variable-frequency **120-volt** source is connected across its terminals,

a) calculate the **resonant frequency**

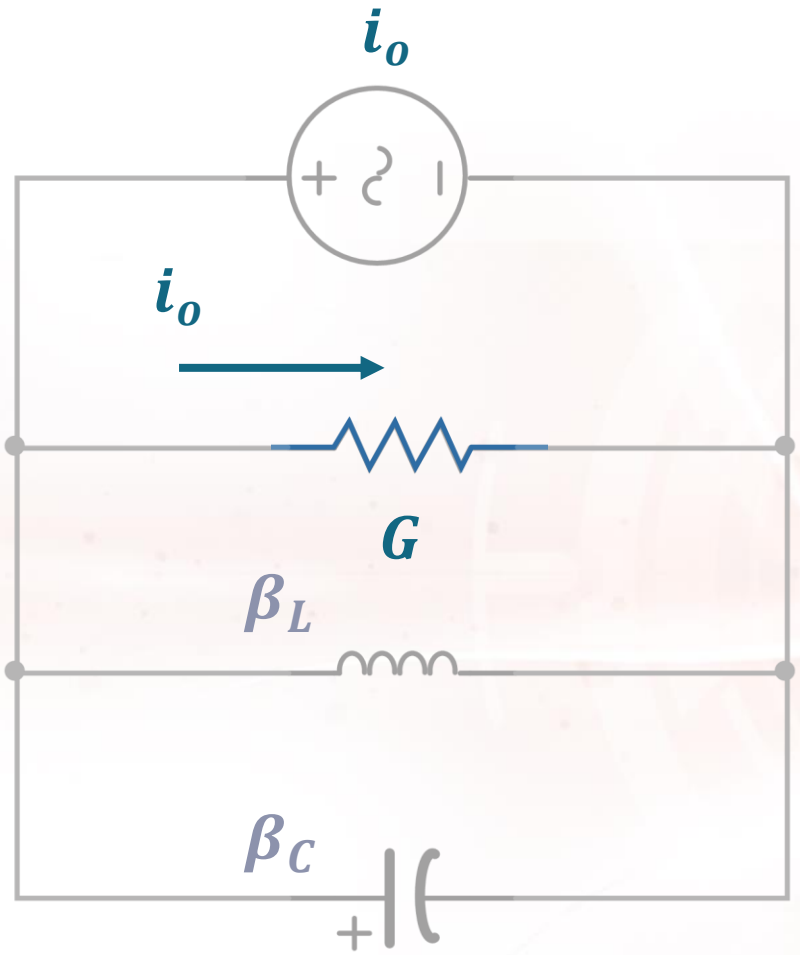
For the condition of resonance,

b) determine the circuit **current** and power

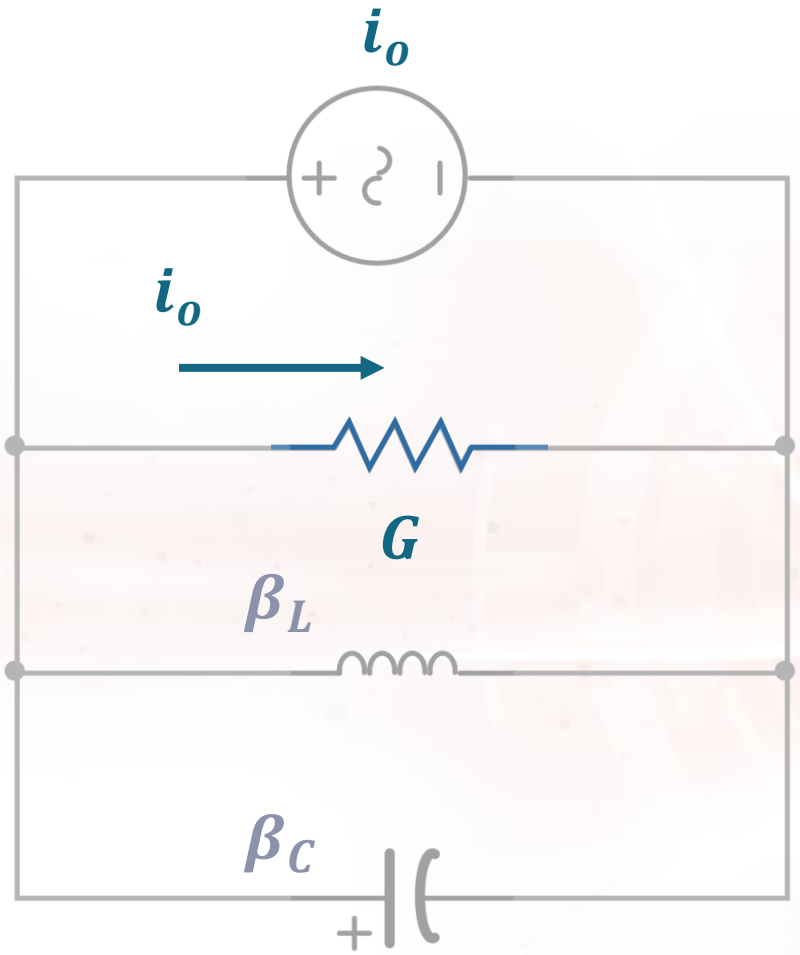
c) the **voltage** drop across the resistor, the inductor, and the capacitor

PARALLEL RESONANCE

$$\beta_L = \beta_C$$



Circuit Diagram



Circuit Diagram

$\beta_L = \beta_C$

Admittance Phasor Diagram

A parallel circuit consists of a **10-ohm** resistor, a **2-henry** inductor, and a **50-μf** capacitor. If a variable-frequency **1-volt** source is connected across its terminals,

a) calculate the **resonant frequency**

For the condition of resonance,

b) determine the circuit **current** and **power**

c) the **voltage** drop across the resistor, the inductor, and the capacitor

LABORATORY